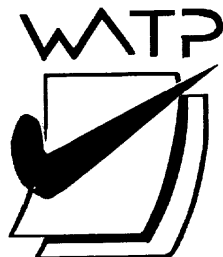


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SEMESTER ONE

MATHEMATICS APPLICATIONS UNIT 1

2017

SOLUTIONS

Calculator-free Solutions

1. (a) 20% of \$8 000 = \$1 600 ✓✓
 (b) 100% is \$80 000 ✓✓
[4]

2. $(17.1 + 16.9)(17.1 - 16.9) = 6.8$ ✓✓
[2]

3. Concession = \$15 ✓
 Discount = \$5 ✓
 $\frac{5}{20} = 25\%$ ✓
[3]

4. (a) \$3.15 ✓
 (b) \$31.50 ✓
 (c) \$315 ✓
[3]

5. (a) $\frac{V_1}{V_2} = \frac{5}{3} = \frac{1000}{x}$ ✓
 $5x = 3\,000$
 $x = 600$
 Volume is 600 cm³ ✓
 (b) $s^3 = 1000$ ✓
 $\therefore s = 10 \text{ cm}$ ✓
[4]

6. SI = 1000 x 0.07 x 3 ✓
 = \$210 ✓
[2]

7. (a) \$5 000 ✓
 (b) 1% per month = 12% per year ✓✓
 (c) 36 months = 3 years ✓✓
[5]

8. (a) $\begin{bmatrix} -1 & -4 & -1 \\ -1 & -2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$ ✓

 (b) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ ✓

 (c) $\begin{bmatrix} -2 & -8 & -2 \\ -4 & -6 & 0 \\ -6 & -4 & 0 \end{bmatrix}$ ✓

 (d) $\begin{bmatrix} 0 & 0 & 0 \\ 6 & 9 & 1 \\ 12 & 18 & 2 \end{bmatrix}$ ✓✓
[5]

9. (a) $\begin{bmatrix} 10\,000 & 8\,000 & 5\,000 \\ 8\,000 & 5\,000 & 10\,000 \end{bmatrix}$ ✓✓
- (b) $[(18\,000) (13\,000) (15\,000)]$ ✓
- (c) $(18\,000 \ 13\,000 \ 15\,000) \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ ✓✓
- $= [135\,000]$
- Total value is \$135 000 ✓ [6]
10. (a) $P = 2(5+4) = 18 \text{ cm}$ ✓✓
- (b) $\text{Area} = (1 \times 1) + (4 \times 4) = 17 \text{ cm}^2$ ✓✓✓ [5]
11. (a) $x^2 = y^2 + z^2$ ✓
- (b) $z^2 = x^2 + y^2$ ✓
- (c) Test to see if angle M is a right angle, since it is opposite the largest side.
Is $6^2 + 5^2 = 7^2$? No, $\therefore M \neq 90^\circ$ ✓✓ [5]
12. (a) $SA = \frac{1}{2}(4\pi(3^2)) + \pi(3^2) = 27\pi \text{ m}^2$ ✓✓
- (b) $V = \frac{1}{2} \left(\frac{4}{3}\pi(3^3) \right) = 18\pi \text{ m}^3$ ✓✓ [4]
13. (a) and (c) ✓✓ [2]

Calculator-assumed Solutions

14. (a) $\angle A = \angle E$ and $\angle B = \angle D$ alternate angles ✓
 $\therefore \triangle ABC \sim \triangle EDC$ (A.A.) ✓
 (b) $\frac{AB}{ED} = \frac{AC}{EC} = \frac{1}{2}$ ✓
 $\frac{4}{EC} = \frac{1}{2}$
 $\therefore EC = 8$ ✓
 $\therefore AE = 12$ ✓ [5]
15. (a) $1.08 \times 6 = 6.48$ m ✓✓
 (b) Extra amount = 1 m ✓
 $1:12 = 8.3\% > 8\%$ ✓
 $\therefore$ Claim is true ✓
 (c) Maximum length paid for is $\frac{6}{1.08}$ metres ✓✓
 $\therefore$ Maximum amount paid = \$16.67 ✓ [8]
16. (a) A: $25\,000 (1.05) = \$26\,250$
 Interest = \$1 250 ✓
 B: $25\,000 (1 + \frac{0.045}{12})^{12} = \$26\,148.50$
 Interest = \$1 148.50 ✓
 C: $25\,000 (1 + \frac{0.048}{2})^2 = \$26\,214.40$
 Interest = \$1 214.40 ✓
 $\therefore$ Best Option is Bank A ✓
 (b) SI = $25\,000 \times 0.05 \times 1 = \$1\,250$ ✓
 So, in this case SI and CI are the same, ✓
 because the period is 1 year. ✓
 (c) $P (1 + \frac{0.045}{12})^{12} = P + 10\,000$ ✓
 $P = \$217\,676$ ✓ [8]
17. (a) Taxable income = \$80 000 ✓
 (b) (i) Tax on \$100 000 = Tax on \$80 000 + Tax on \$20 000 ✓
 $= 17\,547 + 0.37 \times 20\,000$
 $= 17\,547 + 1850$
 $= \$24\,947$ ✓
 (ii) $26\,000 - 24\,947 = \$1\,053$ ✓✓
 (iii) Carolyn would pay half of \$24 947 ✓
 ie \$12 473.50 ✓
 That is $12\,473.50/100\,000 = 12.4735\%$ ✓✓ [9]
18. (a) $30 \times 25 = \$750$ ✓
 (b) $375/10 = \$37.50$ per hour ✓
 (c) $400/50 = 8$ hours ✓
 (d) (i) Prepare financially for retirement ✓
 (ii) 9.5% of 1 525 ✓
 $= \$144.88$ ✓ [6]

19. (a) 2 000 m ✓
 (b) 2.5 cm = 25 mm ✓✓
 (c) Side of square = 2 cm ✓
 $\therefore$ Diagonal of square = $2\sqrt{2}$ ✓
 $\therefore$ Diagonal of square in real life = $2\sqrt{2} \times 400$ m ✓
 = 1 131.37 m ✓ [7]

20. (a)

		To		
(b)		A	B	C
	From A	0	2	2
	From B	2	0	0
	From C	2	2	0
	From B	0	4	4
	From C	2	2	2

There are 6 different ways of getting from A to A in 2 steps.

Eg A-C-A etc

✓

[6]

21. (a) $x = 4$ and $y = 7$ ✓
 (b) $x = 7$ and $y = 5$ ✓
 (c) $x = 3$ and $y = 4$ ✓ [3]

22. (a) T_1 has perimeter of 24 cm. T_2 has a perimeter 3 times larger than T_1 .
 $\therefore T_2$ has sides 3 times those of T_1
 $\therefore T_2$ has sides 21, 24 and 27 ✓✓✓
 (b) $A_2 = 9A_1 = 9\sqrt{720}$ ✓ [4]

23. (a) $V = 550000(1.06)^5 = \$ 736\,024$ ✓✓
 (b) $V = 700\,000(0.96)^7 = 526\,013 \approx \$526\,000$ ✓✓
 (c) (i) Solve $500 = 700(0.96)^n$ ✓
 $n = 8.24$ years
 So, during 2018 ✓✓
 (ii) Markets seldom drop continually for such a long period.
 Prices would probably stabilise and then rise over that period. ✓ [8]
24. (a) $P = \frac{115}{360} 10\pi = 10.04 \text{ cm}$ ✓✓
 (b) $A = \left(\frac{115}{360}\right) \pi 5^2 = 25.09 \text{ cm}^2$ ✓✓
 (c) $A_t = 100 A_s = 2\,509 \text{ cm}^2$ ✓ [5]
25. (a) 5 000 000 rupiah ✓
 (b) $283\,000/10\,000 = \$28.30$ ✓✓
 (c) (i) 300 g: 70 r per gram
 750 g: 64 r per gram ✓
 $\therefore$ large jar is better value ✓
 (ii) A large jar is not needed for just a one week stay. ✓ [6]
26. (a) $\frac{120}{12\,000} = 1\% \text{ per month}$ ✓
 $\therefore 12\% \text{ pa}$ ✓
 (b) For month 36, $1\% \text{ of } \$335.12 = 3.35$ ✓
 (c) 3 years ✓
 (d) $12\,000 + 120 - 400 = 11\,720$ ✓
 (e) $335.12 + 3.35 - y = 0$ ✓
 $y = 338.47$ ✓
 (f) $(35 \times 400) + 338.47 + 3\,000 \text{ (Deposit)} = 17\,338.47$ ✓✓✓
 (g) $17\,338.47 - 3\,000 - 12\,000 = 2\,338.47$ ✓✓
 (h) It includes every number in existence, so tells us very little. ✓ [13]
27. (a) $V = \frac{4}{3} \pi (3)^3 = 36 \pi \text{ cm}^3$ ✓
 $SA = 4\pi 3^2 = 36 \pi \text{ cm}^2$ ✓
 $\therefore$ Numerically the same ✓
 (b) If the radius is any value other than 3, then $V \neq SA$ ✓
 Eg If $r = 4$, $V = \frac{256\pi}{3}$ and $SA = 64 \pi$ ✓✓ [6]
28. $SA_1 = \pi \cdot 5^2 + \pi \cdot 5 \cdot 6 = 55 \pi$ ✓✓
 $SA_2 = 2 \pi \cdot 2^2 + \pi(4)(4) = 24 \pi$ ✓✓
 The ratio of the surface areas = 55:24 ✓
 There are no units for a ratio. ✓ [6]